Project 21 · CSAI

Flight Delay Prediction with Deep Learning

Phase 2: MLP with Entity Embeddings



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From Phase 1 to Phase 2

Phase 1 (Machine Learning)

- Predict arrival 15+ min late, before departure
- Leakage-free features, split by date
- Logistic Regression: PR AUC 0.376
- XGBoost: PR AUC 0.391 (best)

Phase 2 (Deep Learning)

- MLP with learned embeddings for carrier, origin and destination
- Same data, features, split and metrics: a fair comparison
- Regularisation, seed stability and an ablation study
- Calibration and a what-if rescheduling analysis

Phase 2 changes only the model, so any difference in results comes from the model itself.

What the deep learning literature tells us

Entity embeddings

Guo & Berkhahn (2016): map each category to a learned vector, as word embeddings do for words (Mikolov et al., 2013)

Regularisation

Dropout (Srivastava et al., 2014), AdamW weight decay (Loshchilov & Hutter, 2019), early stopping (Prechelt, 1998)

Trees are strong on tables

Grinsztajn et al. (2022), Shwartz-Ziv & Armon (2022): boosted trees often match or beat deep nets on tabular data

Calibration

Guo et al. (2017): neural networks are often over-confident; temperature scaling fixes it with one number

Preparing the data for a neural network

Same 327,346 flights, leakage rules, features and date split as Phase 1. Three changes for the network:

Categories to ids

Carrier, origin and destination become integer ids from the training set. Id 0 means 'never seen' (57 test flights).

Standardised numbers

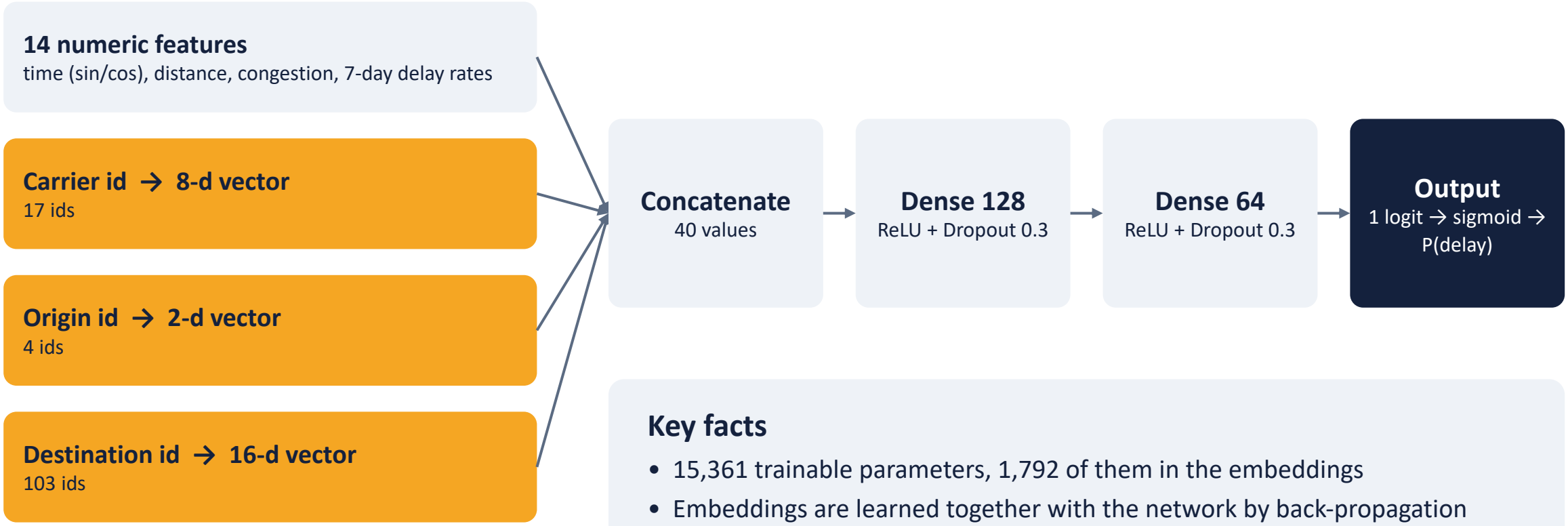
The 14 numeric features are scaled to mean 0 and std 1 with training statistics; missing history gets the training mean.

No class weights

Plain cross-entropy keeps probabilities close to the real delay rate. Ranking (PR/ROC AUC) is barely affected.

Split: train Jan to Aug (217,727) · validation Sep to Oct (55,628) · test Nov to Dec (53,991)

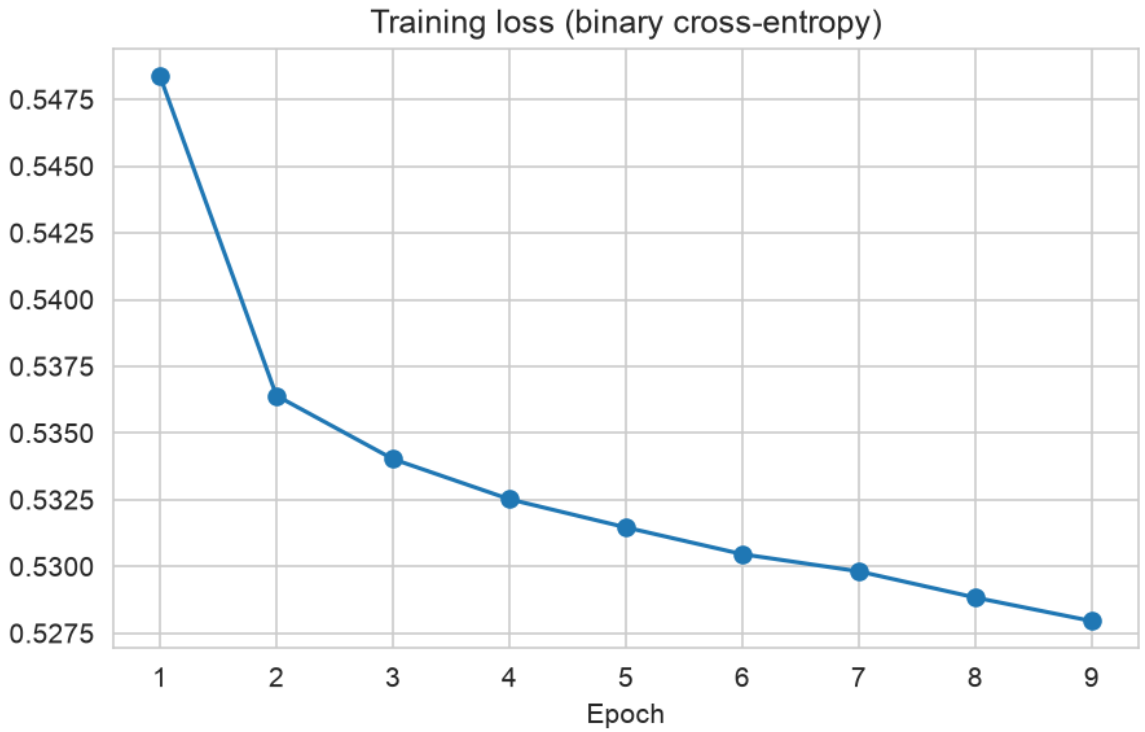
Architecture: MLP with entity embeddings

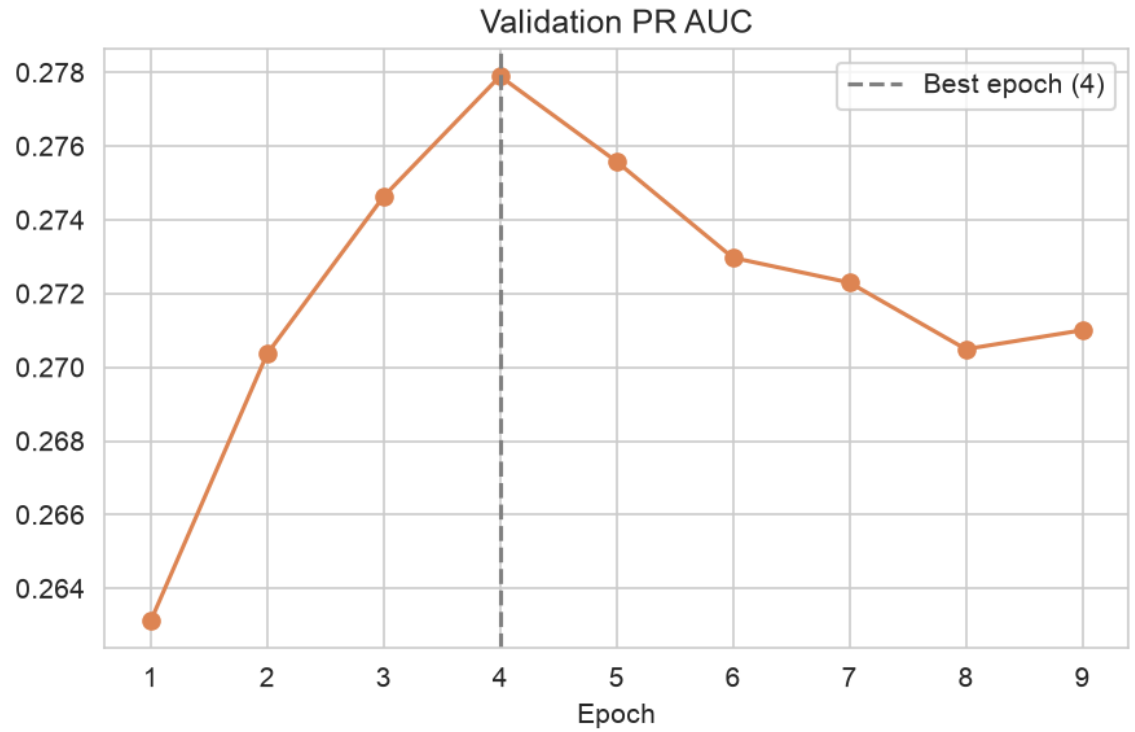


Key facts

- 15,361 trainable parameters, 1,792 of them in the embeddings
- Embeddings are learned together with the network by back-propagation
- Loss: binary cross-entropy on the logit

Training and regularisation





Optimiser

AdamW, learning rate 0.001, mini-batches of 1,024

Regularisation

Dropout 0.3 and weight decay 0.0001

Early stopping

Keep the best validation epoch, stop after 5 without gain

Stable across random seeds

Seed	Epochs	Val PR AUC	Test PR AUC	Test ROC AUC
0	9	0.278	0.386	0.653
1	13	0.273	0.383	0.653
2	8	0.271	0.386	0.649
3	11	0.272	0.380	0.651
4	8	0.271	0.387	0.653

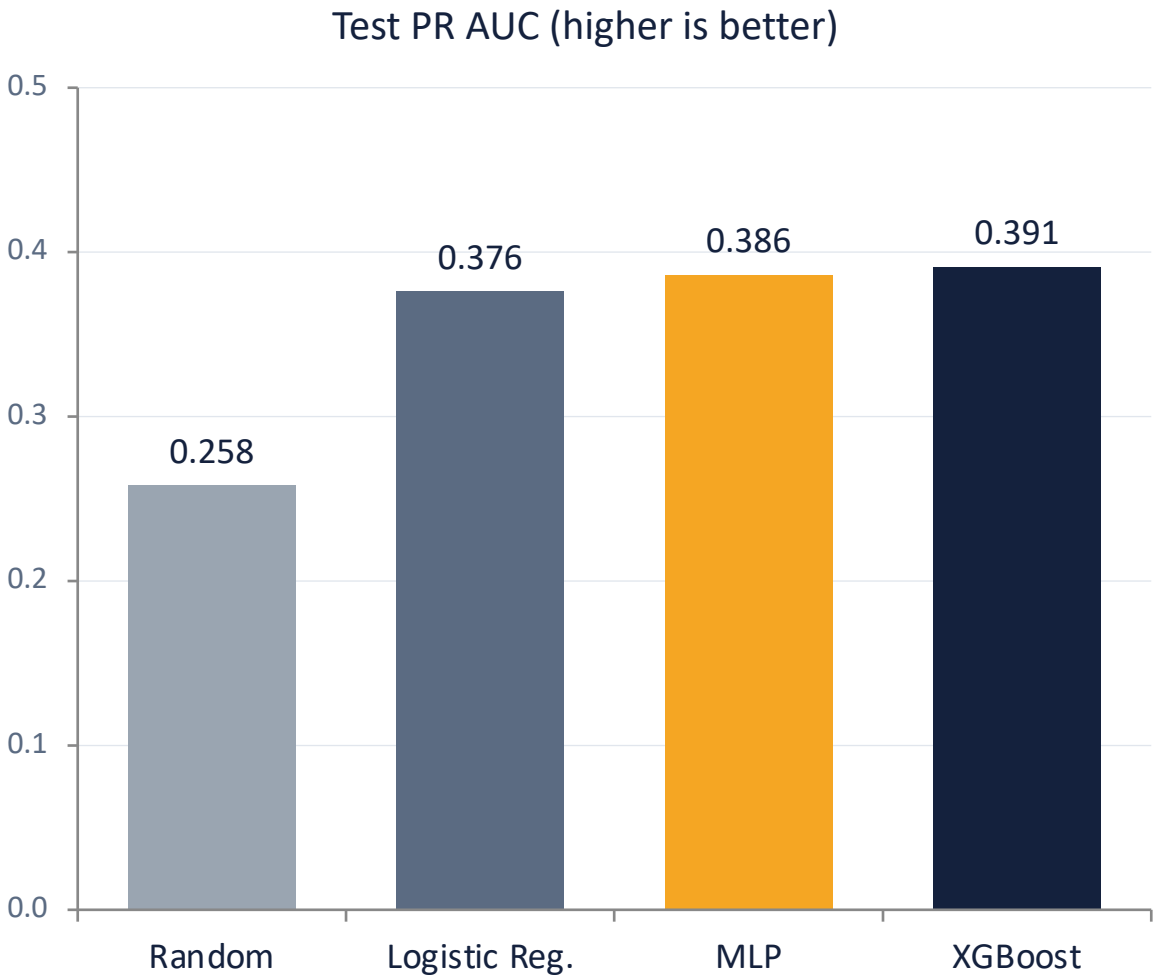
0.385 ± 0.003

test PR AUC over 5 seeds

Random start, batch order and dropout change with the seed

Differences under about 0.005 are noise, which matters when reading the ablation

The MLP lands between the two Phase 1 models

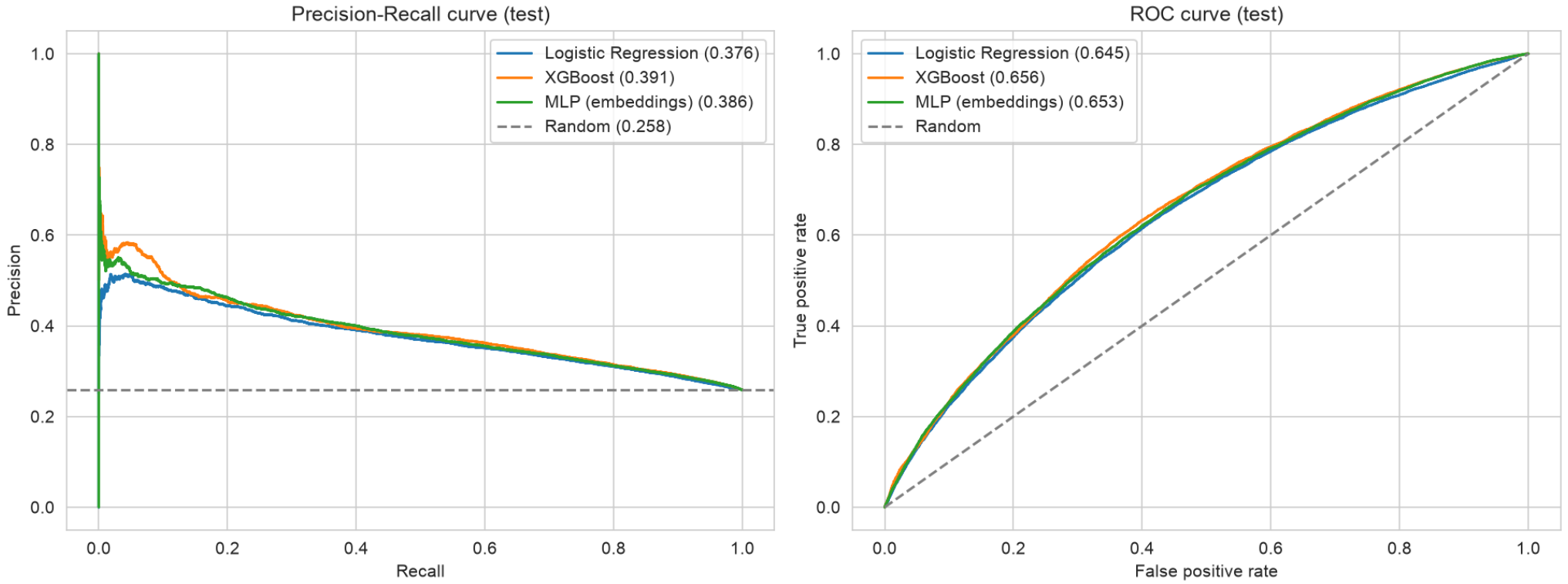


Model	PR AUC	ROC AUC	Rec@P0.5	Brier
Log. Reg.	0.376	0.645	0.059	0.228
XGBoost	0.391	0.656	0.111	0.219
MLP	**0.386**	**0.653**	**0.094**	**0.182**

What it means

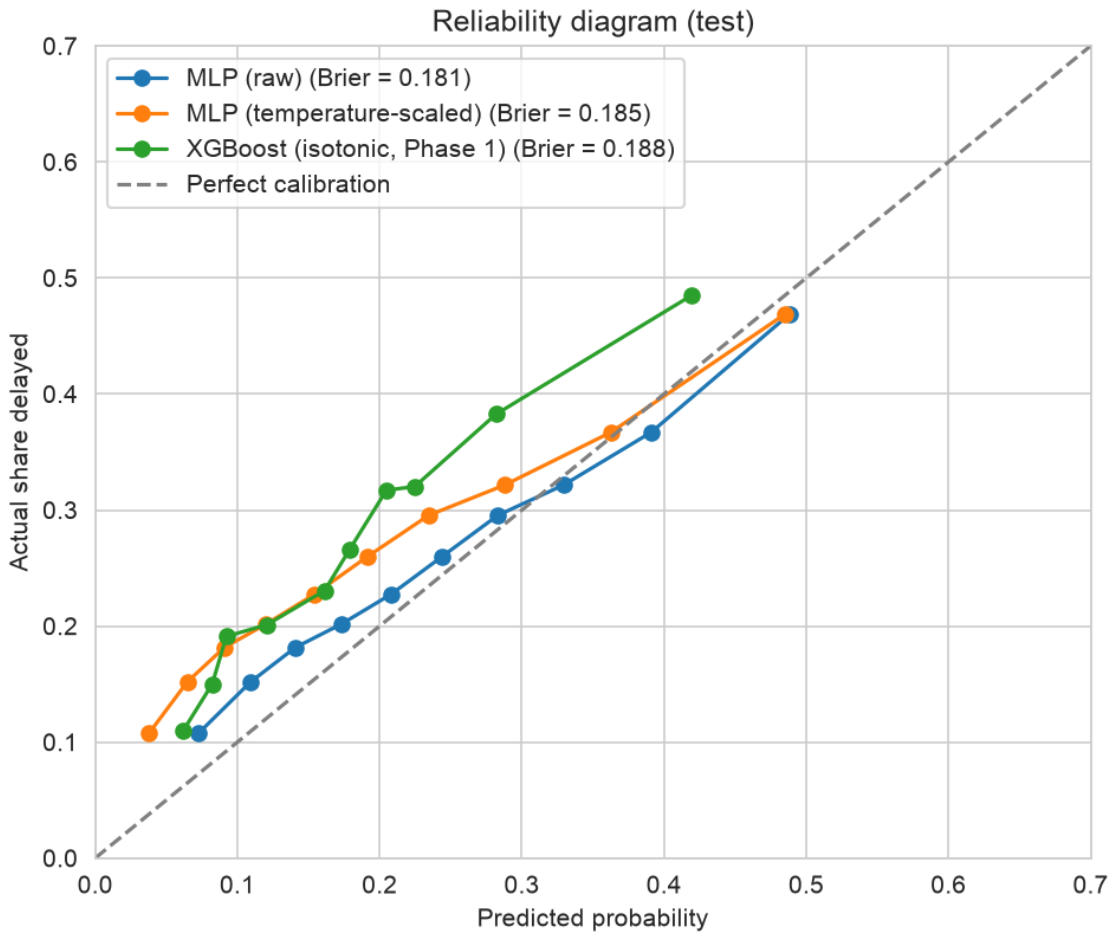
- MLP clearly beats Logistic Regression
- XGBoost still ranks flights slightly better, as the literature expects for tabular data
- MLP has the best probabilities (lowest Brier score)

Precision-recall and ROC curves on the test set



The MLP and XGBoost curves almost overlap; both stay clearly above random guessing. Logistic Regression is a little lower.

The raw MLP gives the best-calibrated probabilities



MLP (raw)
best: no calibration needed

0.182

MLP + temperature scaling
T = 0.785, slightly worse

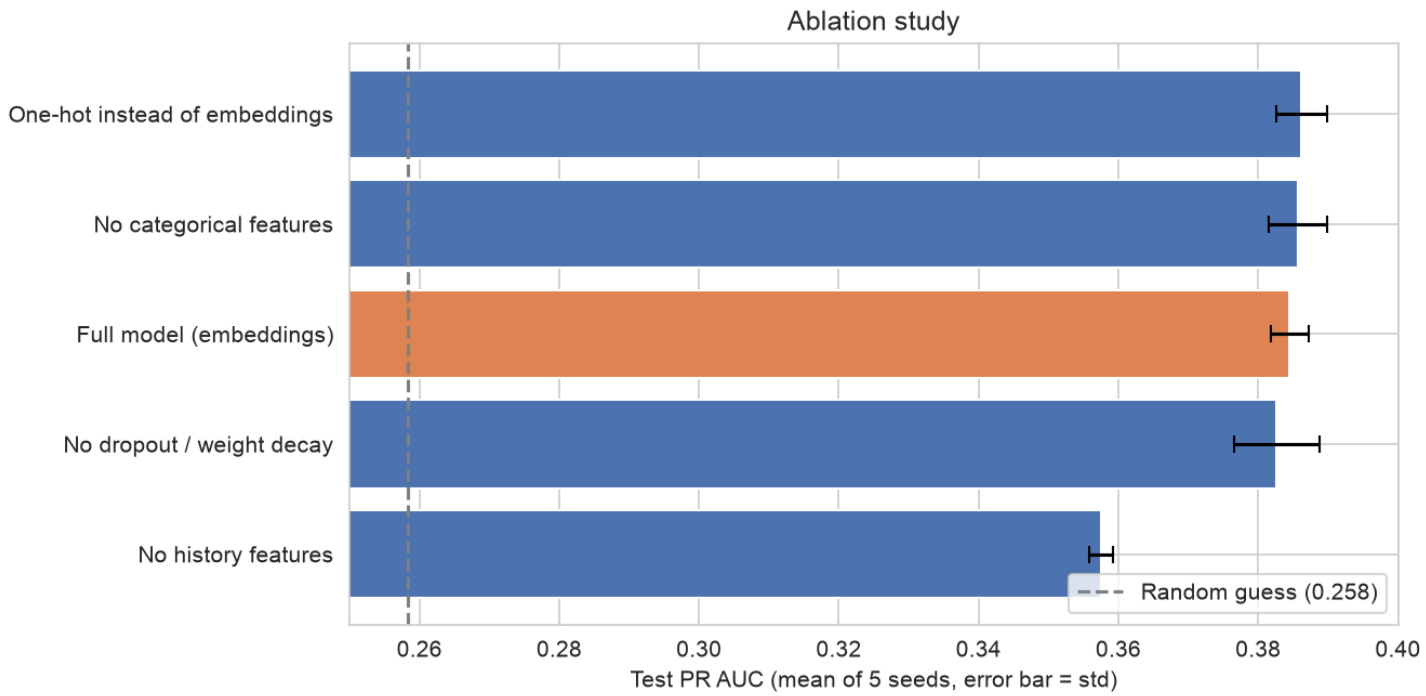
0.185

XGBoost + isotonic (Phase 1)
under-predicts delays

0.188

Brier score, lower is better. Calibration fitted on Sep to Oct (15.7% delayed) pulls test predictions (25.8% delayed) too low.

Ablation: history features matter most



Each variant: 5 seeds; bars show the mean test PR AUC, error bars one standard deviation.

History features

Without them PR AUC drops from 0.385 to below 0.36

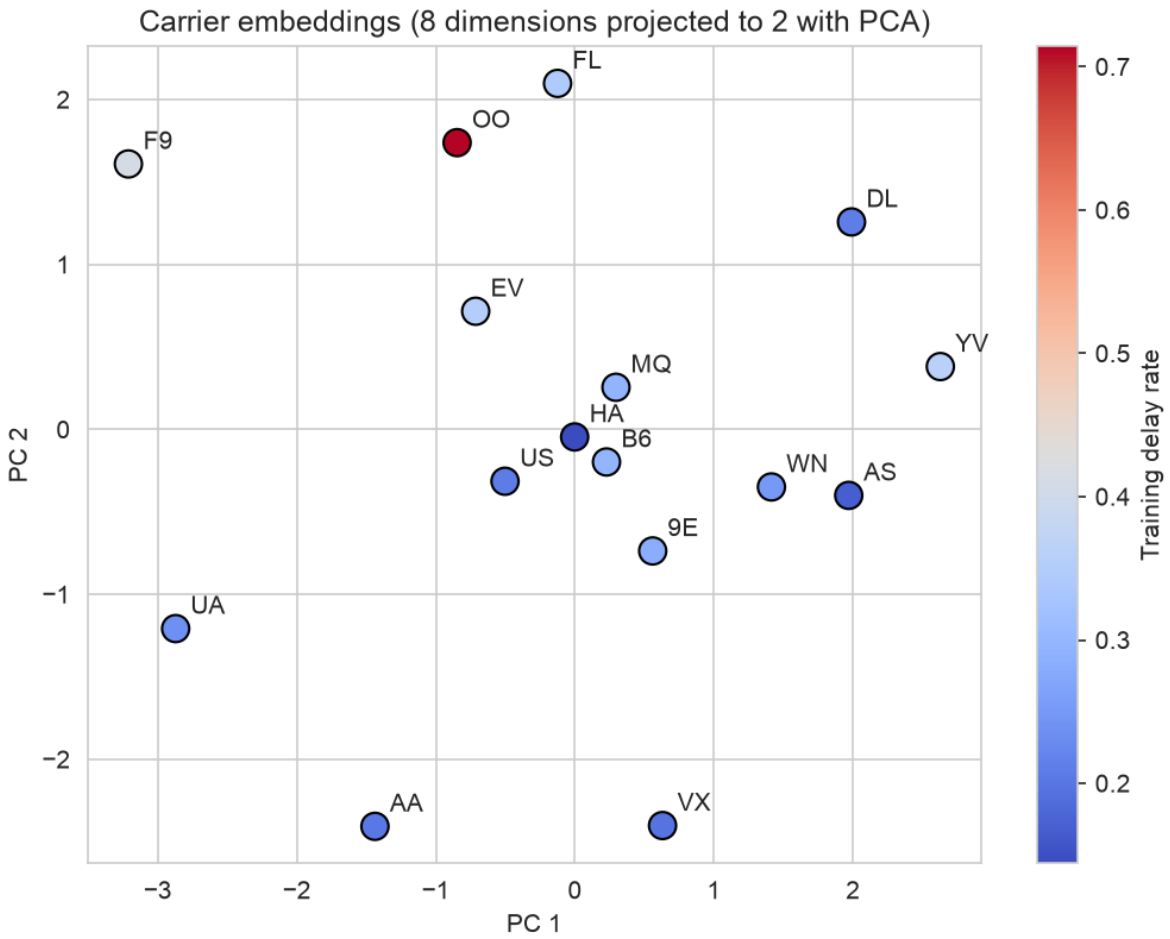
Embeddings

Same as one-hot or no categories at all: the history features already carry that information

Regularisation

Small gain, and half the seed-to-seed spread

What did the carrier embeddings learn?



8 numbers per carrier

Projected to 2 dimensions with PCA for plotting

Weak link to delay rate

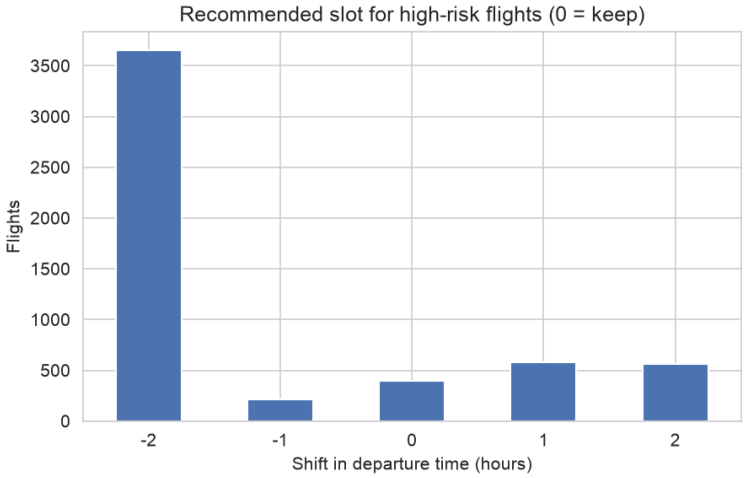
PC 1 correlates only weakly with each carrier's delay rate (r = -0.25)

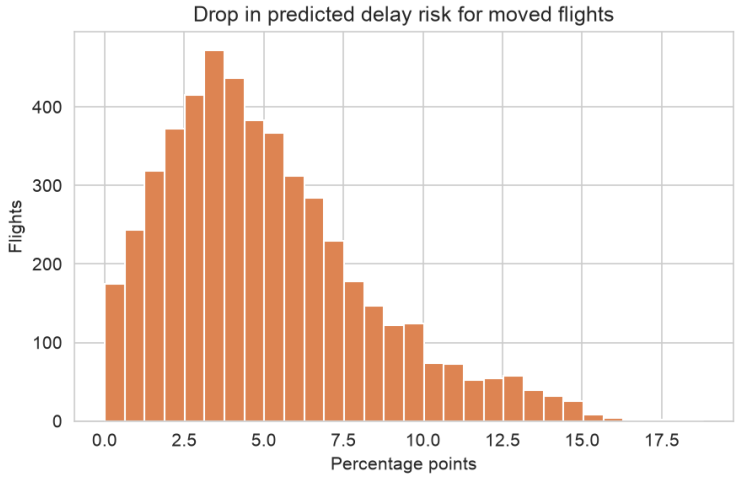
Why

The carrier's recent delay rate is already an input, so the embedding learns smaller differences instead

OO is an outlier: only 7 training flights.

What-if: moving high-risk flights





92.7%
have a lower-risk slot

40.7%
drop 5+ points of risk

0.488 → 0.441
mean predicted risk

Assumptions

- Top 10% riskiest test flights (5,400); move -2 to +2 hours, same day, no arrival after midnight
- Congestion recomputed; route, carrier and history unchanged
- One flight at a time; crews, aircraft and gates ignored. Not causal.

Takeaways and limitations

What we found

- MLP: PR AUC 0.386, between LR (0.376) and XGBoost (0.391)
- Best-calibrated probabilities (Brier 0.182)
- History features drive the predictions; embeddings add little

Limitations

- No weather or air traffic data
- One city, one year; calmer validation months
- What-if is a model estimate, not causal

Next steps

- Weather forecasts as inputs
- Sequence model over ordered aircraft or airport history
- Hybrid: learned embeddings fed to XGBoost

Thank you



Questions?

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github.com/mayank-pillai-99/flight-delay-prediction